



Department of Mathematics and Natural Sciences
Physics -112 Lab Assignment: 01

Resources:

Link for online lab:

https://phet.colorado.edu/sims/html/coulombs-law/latest/coulombs-law_all.html

List and link for the graph-plotting softwares:

- Desmos (Online): [Desmos | Let's learn together.](#)
- Graph (Offline): [Download | Graph \(padowan.dk\)](#)

Tutorials:

- Tutorial link for plotting in Desmos :

<https://www.youtube.com/watch?v=-IIUNWVKnUY>

- Tutorials link for the Graph :

How to install graph software:

<https://youtu.be/e19JqLJMx3A>

How to draw a curve using graph software:

https://youtu.be/QBkdzU_8vVo

How to calculate the slope of a line using graph software:

<https://youtu.be/z4cMiUFu5j8>

Tasks:

This activity consists of two Parts:

1. Part one: Electric force versus distance.
2. Part two: Electric forces versus charge.

Objectives:

1. Satisfy Coulomb's law experimentally
2. Study the parameters that affect the electric force (distance and charge).
3. Find experimentally the electric constant k.

Theoretical Background:

Coulomb's Law: "The magnitude of the electric force that a particle exerts on another is directly proportional to the product of their charges and inversely proportional to the square of the distance between them." Mathematically, the magnitude of this electrostatic force F_E acting on two charged particles (q_1 , q_2) is expressed as:

$$F_E = kq_1q_2/r^2$$

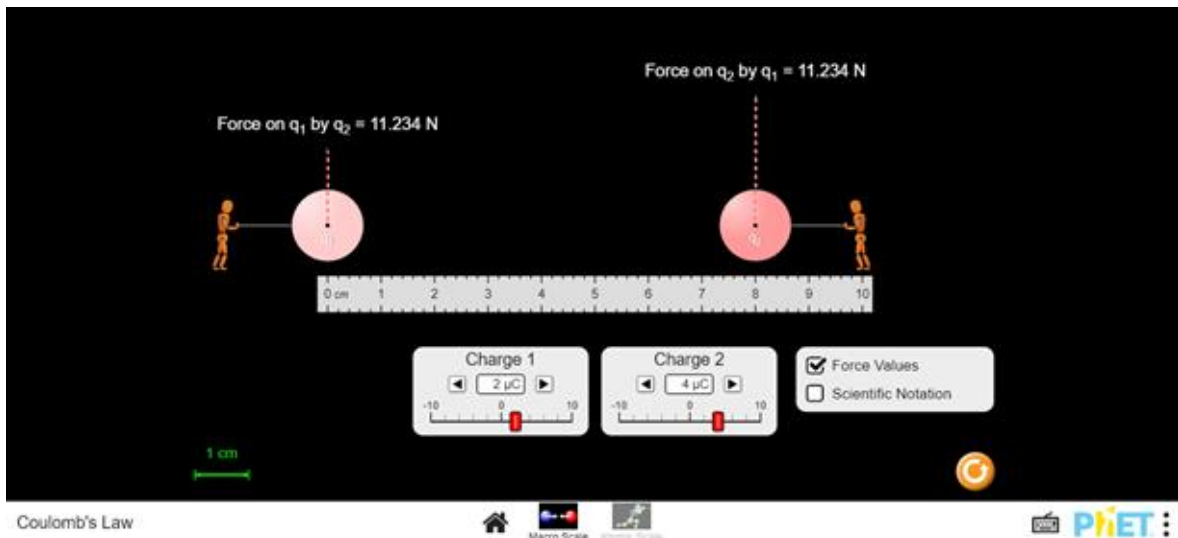
Where r is the separation distance between the charged objects and k is a constant of proportionality, called the Coulomb constant, $k = 9.0 \times 10^9 \text{ Nm}^2/\text{C}^2$.

Part one:

To satisfy the objectives do the following steps.

1. Click on the following link and fix the charge q_1 and q_2 write their values in table.1.

https://phet.colorado.edu/sims/html/coulombs-law/latest/coulombs-law_en.html



2. Change the distance between the two charges as shown in the table.
3. Record the force value for each distance.
4. Fill table 1 by finding r^2 and $1/r^2$.

Table.1 : Data of Force between two charges.

$q_1 = \dots\dots 2\mu C \dots\dots$		$q_2 = \dots\dots 4\mu C \dots\dots$	
r (cm)	r^2 (m ²)	$1/r^2$ (1/m ²)	F_E (N)
10	0.01	100	7.190
9	0.0081	123.46	8.877
8	0.0064	156.25	11.234
7	0.0049	204.08	14.674
6	0.0036	277.78	19.972
5	0.0025	400	28.760
4	0.0016	625	44.938
3	0.0009	1111.11	79.889

Part two:

To satisfy the objectives do the following steps.

1. Click on the following link and fix the charge q_1 and the distance r , write their values in the table https://phet.colorado.edu/sims/html/coulombs-law/latest/coulombs-law_en.html
2. Control q_1 and fix it at $5\mu\text{C}$ and fix the distance between the two objects at 6 cm, record them in table 2.
3. Change the charge of object 2 as shown in table.2 and for each q_2 record the electric force between the two objects in table 2.

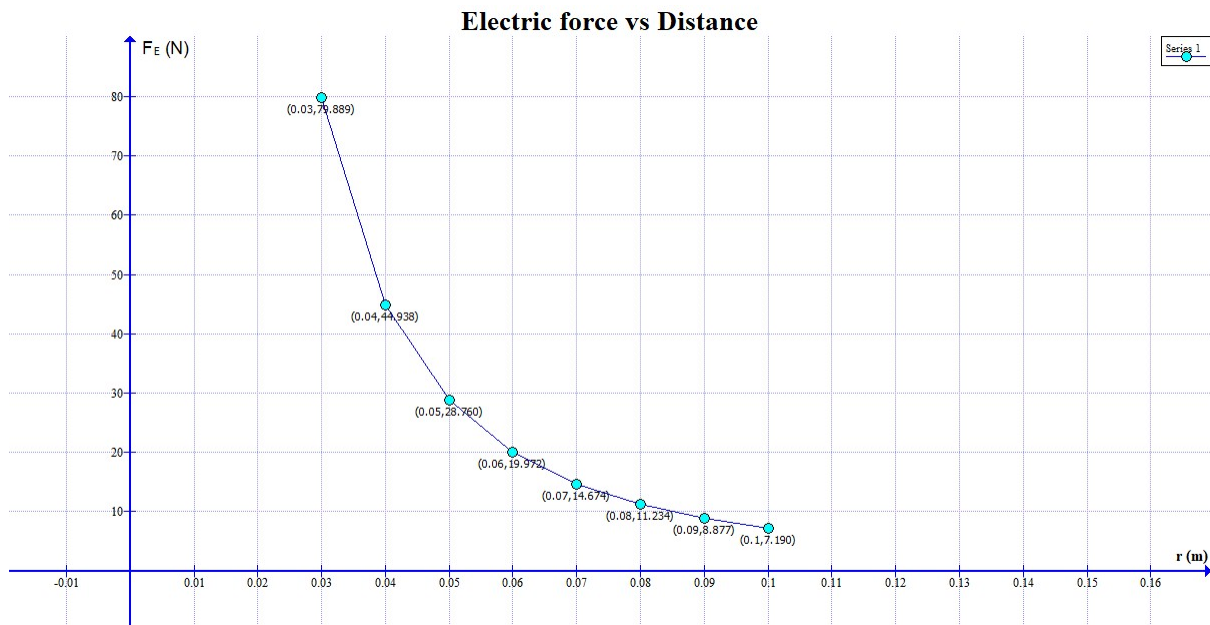
Table.2 : Data for electric force for different charges.

$q_1 = 5\ \mu\text{C}$	$r = 6\ \text{cm}$
$q_2\ (\mu\text{C})$	$F_E\ (\text{N})$
10	124.827
9	112.344
8	99.862
7	87.379
6	74.896
5	62.414
4	49.931
3	37.448

Calculations and Analysis:

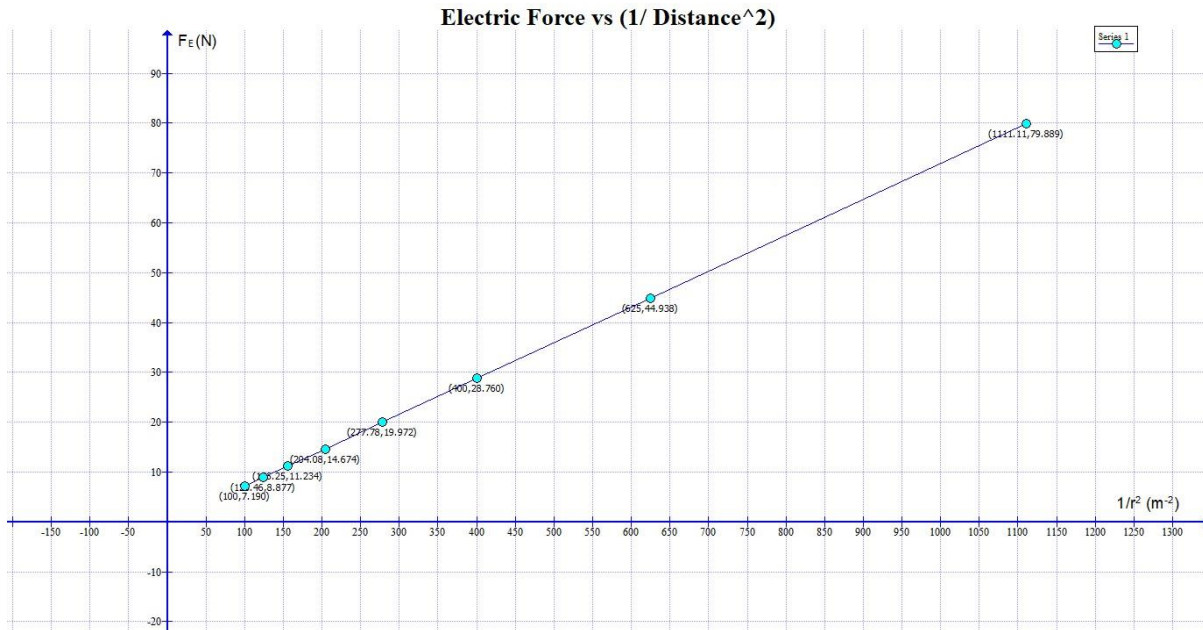
Part one:

1. Plot a graph relating F_E and r . comment on the graph.



The graph shows an exponential curve. As the distance between the charges decreases, the repulsive force between them increases exponentially.

- Plot one more graph relating F_E and $1/r^2$. Use the graph to find the electric constant k .



As we can see in the graph, the slope(m) equals 0.071899, which corresponds to kq_1q_2 . If we divide the slope with q_1q_2 (which are respectively $2\mu C$ and $4\mu C$), we get the electric constant $k = 8.987375 \times 10^9 \text{ Nm}^2/\text{C}^2$

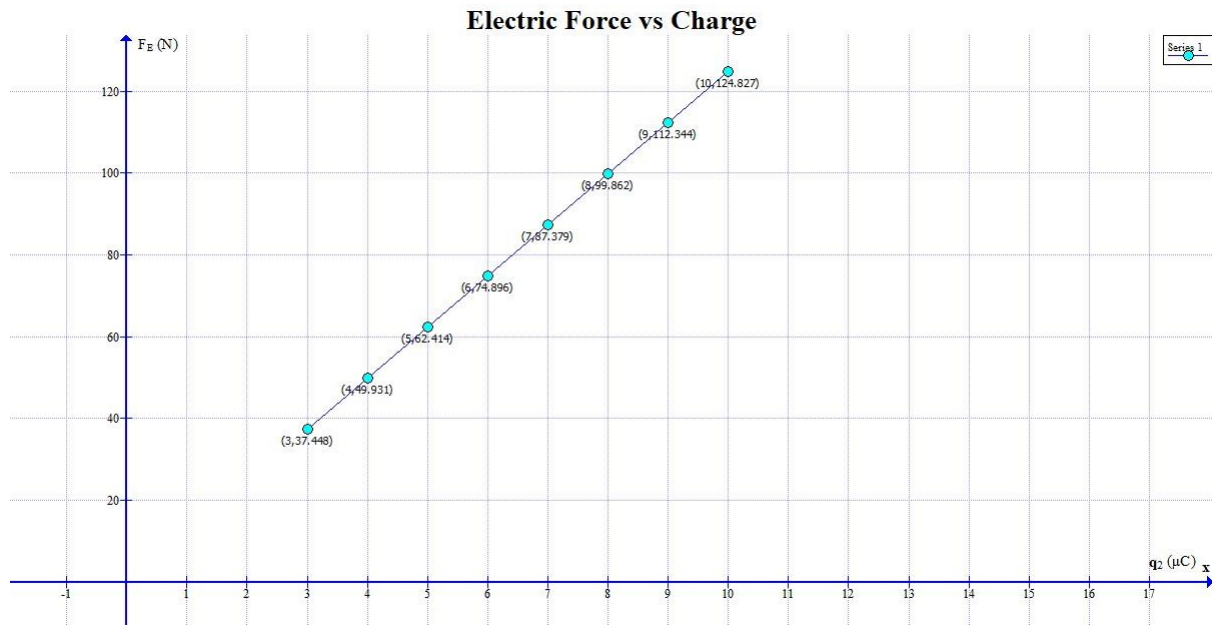
- Calculate the percentage error in k ($k_{\text{known}} = 9.0 \times 10^9 \text{ Nm}^2/\text{C}^2$)

- Percentage of error = $\frac{(8.987375 \times 10^9 - 9.0 \times 10^9)}{9 \times 10^9} \times 100\%$
 $= 0.14\%$

Note: upload graph using google form

Part two:

1. Plot a graph relating F_E and q_2 . comments on the graph.



The graph shows a linear relationship between electric force and charge. As the charge (q_2) increases, the electric force (F_E) increases proportionally. This indicates that the electric force is directly proportional to the charge, following a linear trend.

2. Use the graph to find the electric constant k .

- According to the graph, we found out that the slope(m') is 1.2483×10^7 in SI unit which is equal to kq_1/r^2 . So, if we calculate, we get the electric constant $k = 8.987799 \times 10^9 \text{ Nm}^2/\text{C}^2$

3. Calculate the percentage error in **k** ($k_{known}=9.0 \times 10^9 \text{ Nm}^2/\text{C}^2$)

$$\begin{aligned} \text{- Percentage of error} &= ((8.987799 \times 10^9 - 9.0 \times 10^9) / 9 \times 10^9) \times 100\% \\ &= 0.13556\% \end{aligned}$$

Note: upload graph using google form